

SHONTS AGCM, AB, Canada

WMO: 714050

Lat: 53.3325N

Lon: 112.5395W

Elev: 679

StdP: 93.43

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-32.2	-29.0	-35.5	0.1	-32.2	-32.2	0.2	-28.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	28.4	17.1	26.7	16.7	25.1	16.1	19.5	25.5	18.2	24.2	17.1	22.9	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.3	13.4	22.8	16.0	12.3	21.0	14.8	11.4	19.8	58.6	25.5	54.2	24.3	50.8	23.0	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-36.5	31.8	3.4	1.7	-39.0	33.0	-41.0	34.0	-42.9	35.0	-45.4	36.2		
(5)			-36.6	22.1	3.4	1.3	-39.1	23.0	-41.1	23.7	-43.0	24.4	-45.5	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.5	-10.7	-11.8	-5.4	2.8	10.2	14.4	16.5	15.5	10.6	3.7	-5.1	-12.1
	DBStd	12.17	8.65	8.28	7.83	5.28	4.36	2.97	2.61	3.13	4.40	4.79	7.12	8.26
	HDD10.0	3391	643	609	477	222	52	4	0	1	44	202	453	684
	HDD18.3	5816	901	843	735	467	252	121	68	96	232	455	703	942
	CDD10.0	641	0	0	0	4	58	137	201	172	63	6	0	0
	CDD18.3	25	0	0	0	0	1	4	11	8	1	0	0	0
	CDH23.3	778	0	0	0	3	67	104	245	242	116	2	0	0
	CDH26.7	159	0	0	0	0	11	15	51	53	29	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	413	19	11	17	30	44	70	81	54	33	19	19	16
	PrecMax	574	69	23	37	65	118	128	139	103	82	44	77	39
	PrecMin	221	3	1	6	2	7	21	30	11	4	1	1	3
	PrecStd	73	15	6	8	16	22	28	27	19	21	11	16	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	6.2	12.4	22.4	28.4	28.5	30.7	30.8	30.3	22.6	13.4	6.0
		MCWB	3.0	3.0	6.4	10.6	13.8	16.9	19.0	17.8	16.4	12.2	7.3	2.3
	2%	DB	4.8	4.0	8.9	18.5	25.1	26.1	27.9	28.0	26.6	17.5	8.8	3.8
		MCWB	2.0	1.3	4.4	8.9	12.6	16.0	18.8	17.0	14.8	9.9	4.5	1.0
	5%	DB	3.2	2.2	6.1	14.9	22.6	24.1	26.1	26.0	23.7	14.5	6.0	1.6
		MCWB	0.9	0.3	3.1	7.0	11.7	15.1	18.1	17.2	13.9	8.0	2.7	-0.6
	10%	DB	1.5	0.5	3.9	12.1	20.1	22.2	24.2	24.2	20.3	11.8	3.9	-0.6
		MCWB	-0.4	-1.0	1.3	5.5	10.5	14.0	17.0	16.6	12.5	6.7	1.6	-2.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.2	7.2	11.1	14.7	19.0	21.7	21.0	17.0	12.8	7.8	2.6
		MCDB	6.4	6.0	11.4	22.0	24.3	25.5	27.5	25.4	25.7	21.0	13.0	5.5
	2%	WB	2.2	1.6	4.8	9.0	13.4	17.3	19.9	19.3	15.6	10.5	5.0	1.1
		MCDB	4.5	3.6	8.5	17.8	22.7	23.4	25.9	25.2	25.3	16.1	8.2	3.6
	5%	WB	1.1	0.4	3.0	7.4	12.5	16.1	18.7	18.0	14.2	8.7	3.2	-0.4
		MCDB	3.1	2.0	6.1	14.1	21.0	21.8	24.5	24.1	22.4	13.6	5.6	1.6
	10%	WB	-0.3	-1.0	1.5	5.9	11.3	15.0	17.5	16.8	13.0	7.2	1.7	-2.3
		MCDB	1.5	0.5	3.7	11.1	18.6	20.8	23.0	22.8	19.4	11.4	3.6	-0.7
Mean Daily Temperature Range	5% DB	MDBR	10.4	10.8	10.5	12.1	14.4	12.9	13.3	14.3	14.6	12.3	9.9	9.8
		MCDBR	10.5	11.2	12.1	18.1	18.6	16.3	17.2	18.2	21.4	17.1	12.2	11.6
		MCWBR	8.3	9.1	8.3	9.6	8.3	7.7	9.0	9.2	10.5	10.0	8.6	9.5
	5% WB	MCDBR	10.7	10.9	11.7	16.8	16.8	14.2	15.6	16.1	19.6	16.0	11.8	11.4
		MCWBR	8.8	9.0	8.2	9.3	7.8	7.3	8.9	8.8	10.0	9.6	8.7	9.4
Clear-Sky Solar Irradiance	taub	0.258	0.259	0.307	0.331	0.347	0.361	0.359	0.369	0.311	0.279	0.253	0.246	
	taud	2.303	2.369	2.326	2.363	2.372	2.357	2.395	2.333	2.478	2.541	2.443	2.276	
	Ebn at Noon	733	867	881	902	901	888	884	850	873	839	752	673	
	Edh at Noon	58	74	98	108	113	116	110	110	84	63	50	49	
All-Sky Solar Radiation	RadAvg	1.00	2.04	3.57	4.68	5.74	6.04	6.21	5.11	3.53	2.05	0.99	0.74	
	RadStd	0.11	0.16	0.27	0.45	0.41	0.39	0.30	0.35	0.37	0.24	0.11	0.07	

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(47) Regional (83 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page